

Flat-Top Pulse Generation Using 3-dB Abrupt Taper Interferometric Pulse Shaper

Michael Nix^{*1}, Scott S.-H. Yam¹.

¹Department of Electrical and Computer Engineering, Queen's University,
Walter Light Hall, Kingston, Ontario, K7L 3N6, Canada

^{*}Corresponding Author: 3mn11@queensu.ca

Abstract: Using a novel and inexpensive 204 mm long 3 dB abrupt taper pulse shaper, a 3.8 ps flat-top pulse with less than 4% ripple was generated from a 2.2 ps Gaussian pulse.

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1. Introduction

Flat-top optical pulses have many applications, and are particularly needed in high-speed optical communications for reducing the timing and jitter sensitivity in the optical demultiplexing of high bit-rate pulse trains [1-3]. Previous work has demonstrated the generation of such flat-top pulses using non-linear polarization rotation [4], as well as through the use of long-period gratings (LPG) [1,5].

Abrupt fiber tapers have been previously used in interferometric experiments as Michelson and Mach-Zehnder refractive-index sensors [6, 7] and operate under similar principles as LPGs. Like LPGs, abrupt fiber tapers couple power from the propagating core mode to the cladding modes and it is possible to construct these tapers such that 3 dB of power remains in the core. A Mach-Zehnder interferometer (MZI) constructed from two cascaded abrupt fiber tapers separated by bare fiber also closely resembles its LPG counterpart where the LPGs are just replaced by tapers as shown in Fig. 1a.

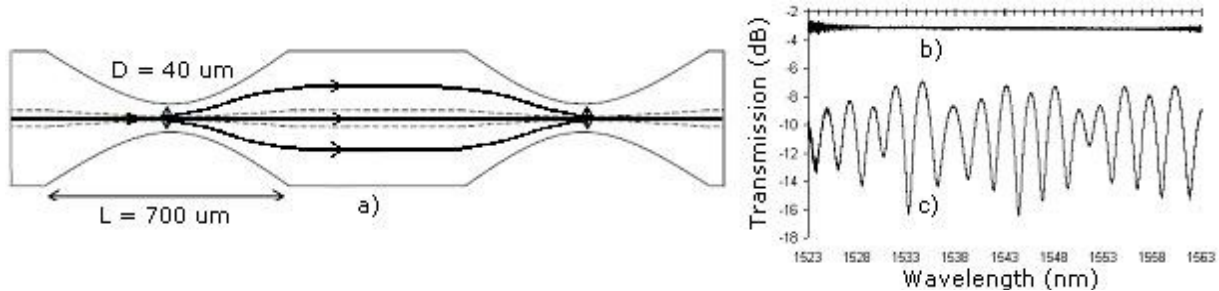


Fig. 1: a) Ideal 3 dB abrupt taper Mach-Zehnder interferometer, b) single taper response, c) MZI response.

Compact, in-line interferometers based on abrupt taper concatenation offer distinct advantages over its LPG-based counterparts. While LPG fabrication requires additional lasers, amplitude masks and exacting photolithographic procedures, abrupt tapers only require an automated programme in commercially available fusion splicers, significantly reducing the production cost. Also, LPG resonant wavelengths are essentially device dependent due to the necessary periodicity of the LPGs' dielectric stack. Tapers, on the other hand, act as simple wavelength independent mode couplers exhibiting a flat wavelength response as seen in Fig. 1b. Another advantage of abrupt taper interferometers is that uniform heating of the fiber by the fusion splicer makes them less sensitive to input polarization. In this paper we present an abrupt taper-based Mach-Zehnder interferometric pulse shaper that is capable of generating 3.8 ps flat-top pulses with less than 4% ripple from a 2.2 ps input Gaussian pulse. This is comparable to current LPG-based pulse shapers that also exhibit a near 2:1 input-to-output pulse-width ratio [1-3,5]. We also present numerical modeling that confirms these results.

2. Interferometer Fabrication

To produce the abrupt tapers, an Ericsson FSU 995 FA fusion splicer was used to heat and draw a piece of stripped single-mode fiber (SMF-28) until the insertion loss measured reached 3 dB. The resulting 3-dB taper has a waist of $40 \mu\text{m}$, a length of $700 \mu\text{m}$ as specified in [8]. To get the appropriate spectral response necessary for filtering of the 2.2 ps Gaussian pulse, two of these tapers were placed 204 mm apart and the chemical coating along that length was

removed such that there was no attenuation of the modes excited in the fiber cladding. The spectral response of the single taper and MZI was measured using a broadband source (BBS) and optical spectrum analyzer. The difference between the original BBS spectrum and the filtered spectrum after a single taper and then the MZI were then subtracted from one another to get the final spectral response as seen in Fig. 1b and 1c.

3. Simulation Setup

The simulation was set up as a 204 mm long piece of coating stripped single mode optical fiber consisting of the original core mode and the excited cladding modes. Each mode has the same Gaussian pulse envelope with a coupling coefficient, a_m , and group delay, τ_m . For the six dominant core and cladding modes, the coupling coefficients were found from numerical simulations of the ideal 3 dB taper [8], and the group delays were computed using a numerical mode-solver of the coating stripped single mode fiber (SMF-28). The spectrum at the end of the interferometer was then computed by Eq. 1:

$$H(f) = \sum_m a_m \exp(-b f^2) \exp(-j 2\pi \tau_m L f) \quad (1)$$

where L is the length of the fiber middle section, and b is a parameter chosen such that the associated time envelope had a full width at half maximum (FWHM) of 2.2 ps. The flat-top pulse was then found by taking the inverse Fourier transform of this spectrum.

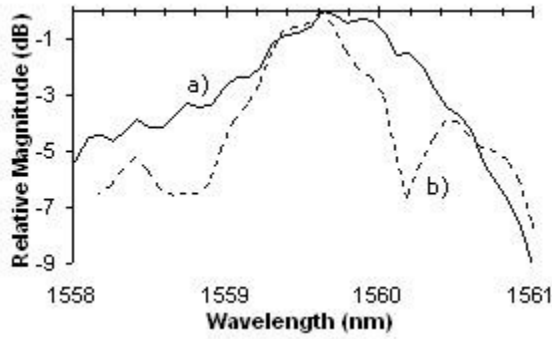


Fig 2: Experimental: Normalized laser spectrum envelope: a) before MZI, b) after.

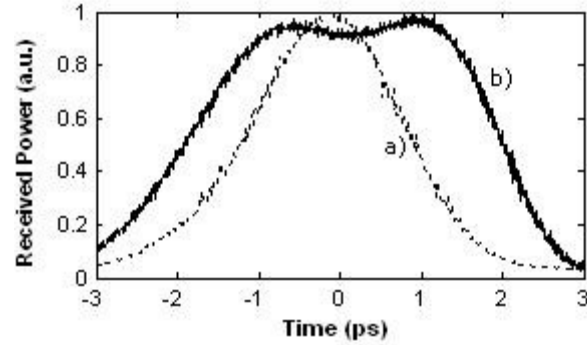


Fig 3: Experimental: Normalized pulse shape: a) before MZI, b) after.

4. Experimental Results and Discussion

The pulse source was a wavelength tunable Pritel RZT mode-locked laser producing almost transform-limited Gaussian pulses with a normalized spectrum envelope as shown in Fig. 2a and 2.2 ps wide as shown in Fig. 3a; the pulses were measured by an Agilent Ultra Wide-Band Optical Sampling Scope (800 GHz bandwidth).

To achieve ideal flat-top rectangular pulses, the output from the interferometric pulse-shaper has to display a sinc function amplitude spectrum. Fig. 1c. shows the experimental output response spectrum, $H(f)$, from the 204 mm long 3-dB abrupt taper interferometric pulse shaper. To obtain a sinc-like spectrum using a near Gaussian input spectrum, the laser peak wavelength should be centered at a transmission peak with adjacent lobes able to properly shape the Gaussian spectrum into that of a sinc-like function. From the interferometer spectrum, there are two areas that might satisfy this requirement: the three lobes centered around 1545 nm, and the area stretching from 1555 – 1565 nm. The best response was found at the peak centered at 1559.75 nm.

Fig. 2b shows the output spectrum envelope as measured from the pulse shaper and it does indeed resemble the necessary sinc function for a flat-top pulse. The temporal pulse as measured by the wide-band sampling scope is shown in Fig. 3b. This pulse has a FWHM of 3.8 ps and ripple of 4%. Fig. 4 shows the spectrum of the simulated results and Fig. 5 shows the resultant pulse. The resultant simulated FWHM pulsewidth is 3.3 ps with a ripple less than 4%. The discrepancy in the pulsewidth between experiment and simulation is due to the simulation assumption of transform limited Gaussian input pulses while the actual pulse source pulses were not transform limited and to the relatively low MZI transmission null depth of 8 dB compared to the 12-30 dB extinction seen in other devices [1-5].

The polarization sensitivity of the MZI pulse shaper was also studied. To do this, the output from the broadband source was linearly polarized and then the polarization rotated while the spectrum was measured. The pulses themselves could not be measured due to the polarization sensitivity of the optical sampling scope. Changing the input polarization through the abrupt taper pulse shaper had no change in the output optical spectrum (Fig. 1c.) showing no polarization sensitivity of the insertion loss through the device.

Despite the input pulses exhibiting a relatively wide spectrum, the effects of chromatic dispersion were not considered through the course of the experiment and simulation as intermodal dispersion was by far the dominant dispersive mechanism in the MZI. For the same frequency content of the input pulse, intermodal dispersion showed a differential mode delay (DMD) of 19.5 ps/m while chromatic dispersion contributed a spread of 0.12 ps/m.

Similar to the LPG realization of MZIs, 3 dB abrupt taper interferometric sensors and pulse shapers can be easily manufactured with different spectral characteristics just by changing the length of the device. For shorter pulses (that exhibit a wider spectrum) the transmission nulls of the frequency response need to be farther apart (e.g. 12 nm for filtering of 900 fs pulses [5]), which can easily be accomplished by shortening the device length.

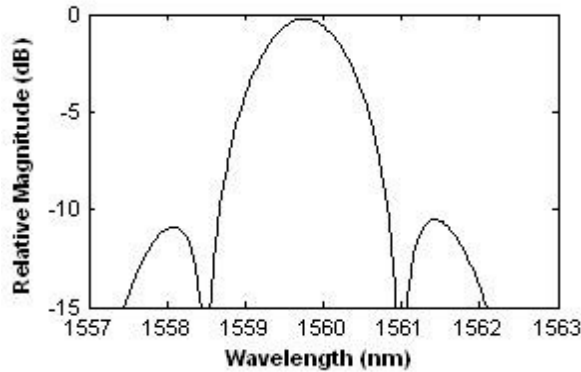


Fig. 4: Simulated Pulse Spectrum.

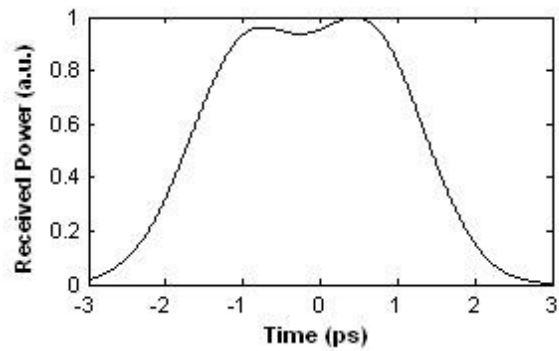


Fig 5: Simulated Pulse.

5. Conclusion

In this paper we have presented an inexpensive, wavelength independent and polarization insensitive 3 dB abrupt taper-based Mach-Zehnder interferometric pulse shaper capable of producing 3.8 ps FWHM flat-top pulses with less than 4% ripple. Simulations based on previous work that determined the 3 dB abrupt taper modal coupling coefficients also confirm these experimental results. For the filtering of shorter pulses, the interferometer as described can easily be modified such that the transmission nulls are farther apart by decreasing the length of the device during fabrication. The 3 dB abrupt taper interferometric pulse shaper is an inexpensive and simple solution for providing flat-top pulses needed in the next generation of high-speed optical communications.

Acknowledgements

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